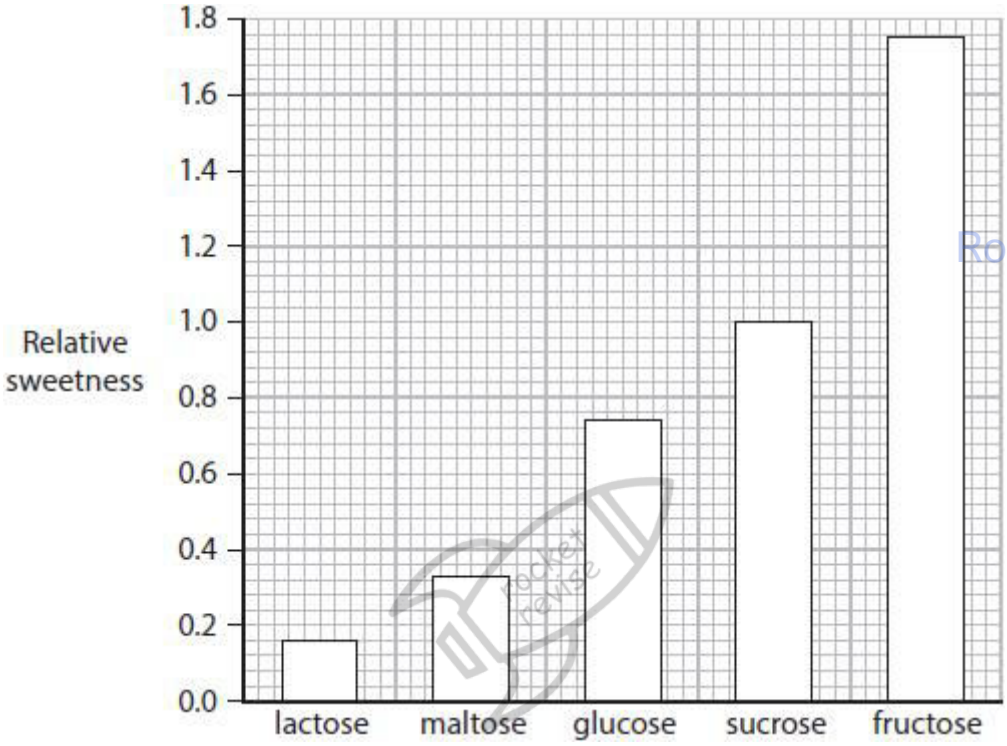
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Questions

Q1.

Carbohydrates are important molecules in living organisms.

(a) The graph shows the relative sweetness of some monosaccharides and some disaccharides.



(i) Which row in the table gives the correct number of monosaccharides and disaccharides shown in this graph?

(1)

	Number of monosaccharides	Number of disaccharides
☐ **A**	1	4
☐ **B**	2	3
☐ **C**	3	2
☐ **D**	4	1

(ii) Which bond joins two monosaccharides together to form a disaccharide?

(1)

- ☐ **A** ester
- ☐ **B** glycosidic
- ☐ **C** hydrogen
- ☐ **D** phosphodiester

(iii) State **two** conclusions that can be made about the relative sweetness of monosaccharides and of disaccharides.

(2)

1 .....

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2 .....

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(iv) Devise an investigation that a student could carry out to confirm the data shown in the graph.

(3)

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(b) Amylose and amylopectin are components of starch.

Compare and contrast the structure of amylose with the structure of amylopectin.

(3)

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(Total for question = 10 marks)

Q2.

Carbohydrates are one of the main types of nutrient. They include polysaccharides, disaccharides and monosaccharides.

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(a) (i) Which formula is correct for a monosaccharide?

(1)

- ☒ **A** $C_nH_nO_{2n}$
- ☒ **B** $C_nH_{2n}O_n$
- ☒ **C** $C_{2n}H_nO_n$
- ☒ **D** $C_{2n}H_{2n}O_n$

(ii) Name the reaction that joins two monosaccharides together to form a disaccharide.

(1)

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(b) The diagram shows some information about carbohydrates.

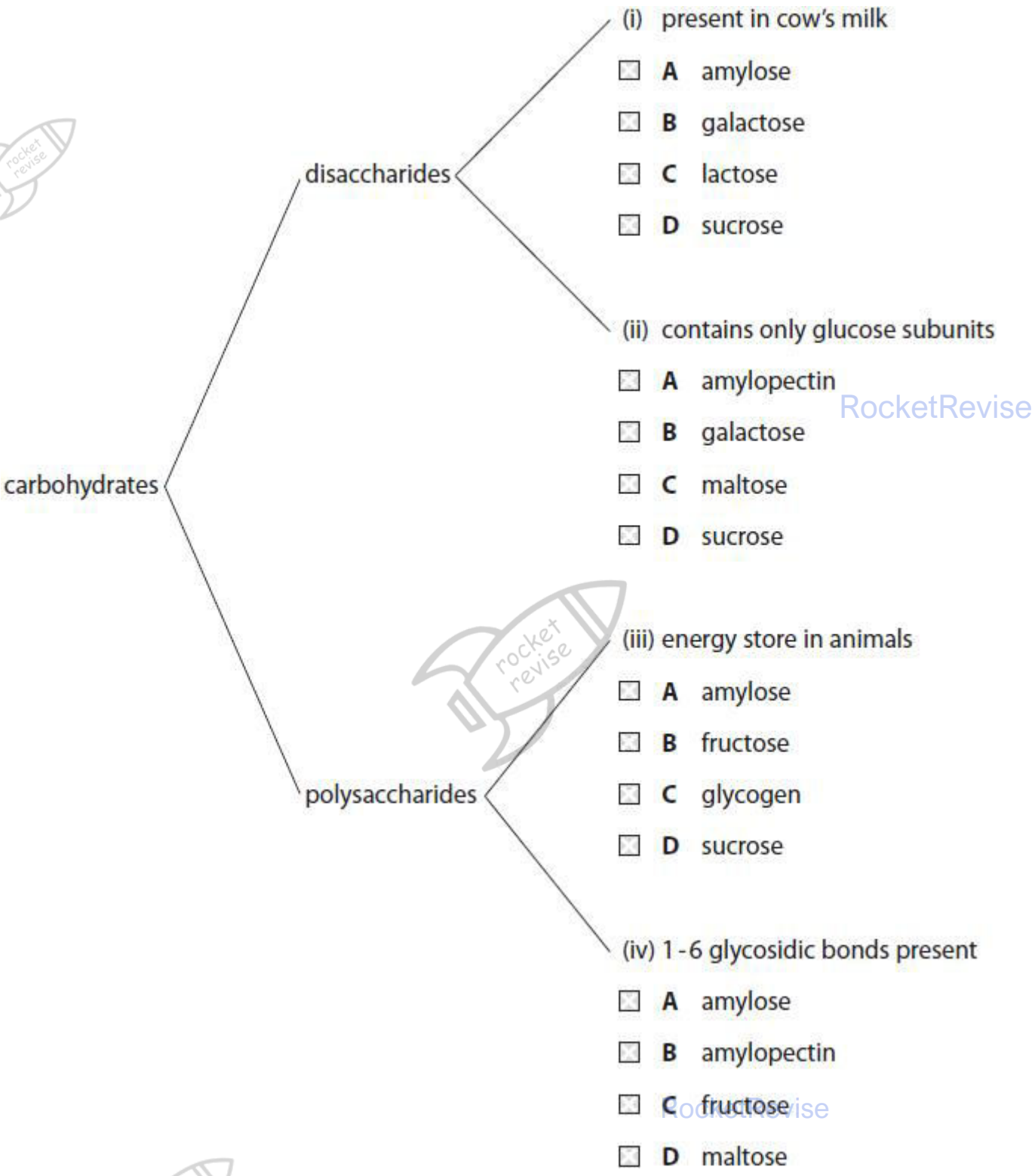
Which carbohydrate is being described in each of (i) to (iv)?

(4)

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(Total for question = 6 marks)

Q3.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

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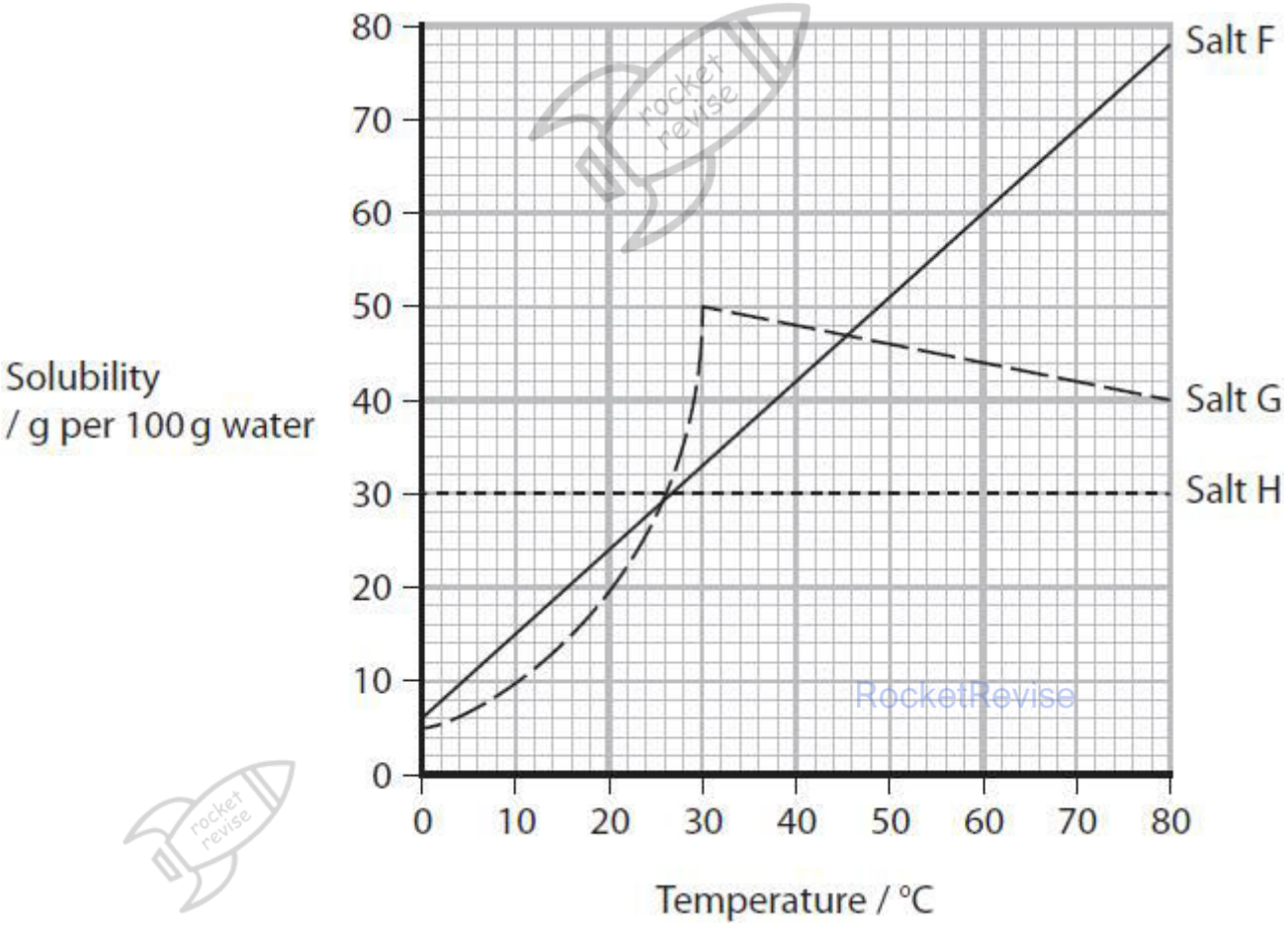
Water is important as a solvent for transport in living organisms.

(a) Draw a diagram of a water molecule to show its dipole nature.

(2)

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(b) The graph shows the effect of temperature on the solubility in water of three salts, F, G and H, in the human diet.



(i) Describe the effect of temperature on the solubility of these three salts.

(3)

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(ii) How many times more soluble is salt G than salt H at 30 °C?

(1)

- ☒ **A** 15.00
- ☒ **B** 1.67
- ☒ **C** 1.50
- ☒ **D** 0.60

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(Total for question = 6 marks)

Q4.

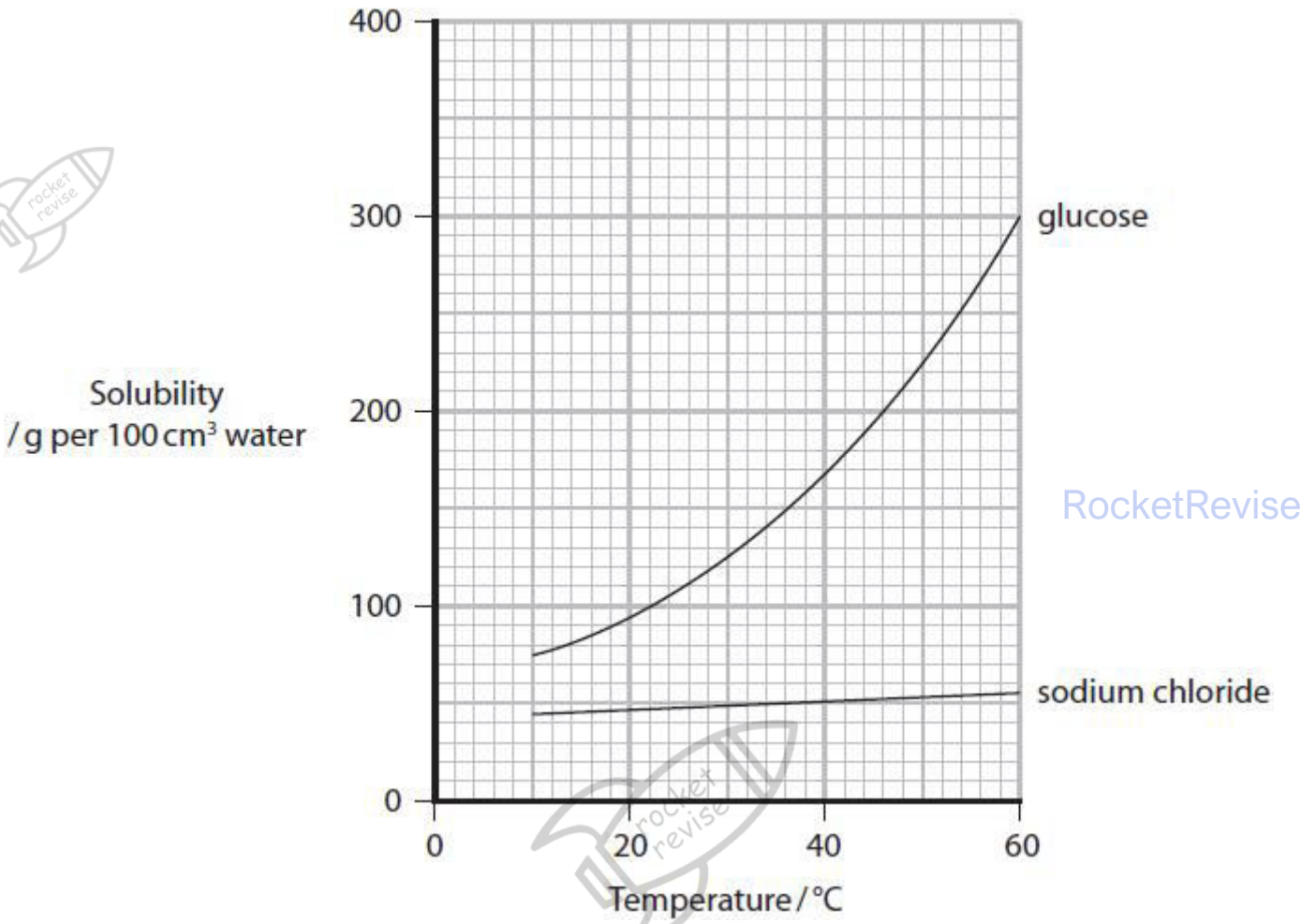
Water is the solvent for the transport of sodium chloride and glucose in the blood.

(a) The graph shows the effect of temperature on the solubility of sodium chloride and glucose in water.

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(i) Compare and contrast the effect of temperature on the solubilities of sodium chloride and glucose in water.

(3)

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(ii) The formula mass of sodium chloride is 58.44.
Calculate the molecular mass of glucose, using the information in the table.

(2)

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Element	Atomic mass
carbon	12
hydrogen	1
oxygen	16

Answer .....

(iii) Calculate how many times greater the molecular mass of glucose is than the formula mass of sodium chloride.

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(1)

Answer .....

(iv) Suggest why there is a difference in the solubility of sodium chloride and glucose in water.

(2)

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(b) Explain why fatty acids are less soluble in blood than glucose and sodium chloride.

(2)

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(Total for question = 10 marks)

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Q5.

Water is an important biological molecule.

Read through the following description of water.

Complete the description by writing the most appropriate word on the dotted lines.

Water has an uneven charge distribution so it is described as a

..... molecule. The hydrogen ends of the molecule have a very slightly ..... charge.

Water is involved in the transport of substances so it is an important

..... in living organisms.

Water is needed in chemical reactions called .....

reactions that break down disaccharides such as

..... into glucose and galactose.

(Total for question = 5 marks)

Q6.

Some questions must be answered with a cross in a box ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

The photograph shows a saffron crocus plant.

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(Source: © NatureOnline / Alamy Stock Photo)

(a) This plant grows from a bulb.

The bulb contains starch as an energy storage molecule for the crocus.
How many of the following statements are correct for starch in living cells?

- starch consists of a mixture of two types of polysaccharide
- starch contains only 1–6 bonds
- starch is insoluble in water

(1)

- ☒ **A** none
- ☒ **B** one
- ☒ **C** two
- ☒ **D** three

(b) Saffron is a spice that is derived from parts of the flower of the saffron crocus.

Saffron contains a disaccharide called gentiobiose.

(i) Read through the following description of some disaccharides.
Complete the description by writing the most appropriate word on the dotted lines.

(4)

Disaccharides consist of two monosaccharides joined together by a
..... covalent bond.
Sucrose and lactose are both disaccharides. They both contain a molecule of
..... . Sucrose also contains one molecule of
..... and lactose also contains one molecule of
..... .

(ii) Gentiobiose is formed from two identical monosaccharides.
Name the type of reaction that joins these monosaccharides together in the formation of gentiobiose.

(1)

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(iii) The molecular mass of each of the monosaccharides in gentiobiose is 180.
The table shows the molecular mass of the elements present in the monosaccharides.

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Element	Molecular mass
carbon	12
hydrogen	1
oxygen	16

Which is the molecular mass of gentiobiose?

(1)

- ☒ **A** 144
- ☒ **B** 162
- ☒ **C** 342
- ☒ **D** 360

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(Total for question = 7 marks)

Q7.

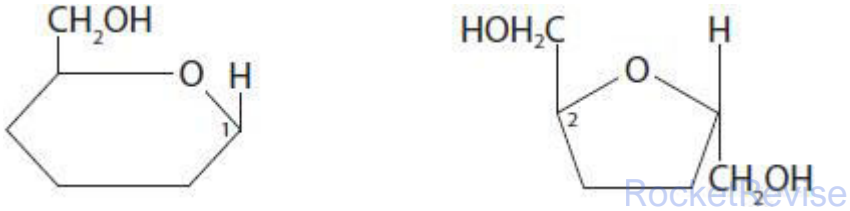
Sucrose is a disaccharide made from glucose and fructose.

Glucose is joined to fructose by a 1–2 glycosidic bond.

(a) (i) The diagram shows a molecule of glucose and a molecule of fructose.

Carbon 1 on the glucose and carbon 2 on the fructose are numbered.
Complete the diagram to show a glycosidic bond between the glucose molecule and the fructose molecule.

(1)



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(ii) Complete the formula equation for the reaction that makes sucrose from glucose and fructose.

(2)

$C_6H_{12}O_6 + C_6H_{12}O_6 \longrightarrow$

(b) The concentration of sucrose in a solution can be determined by a test.

The chart shows the steps involved in this test.

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Break sucrose down into glucose and fructose using an enzyme

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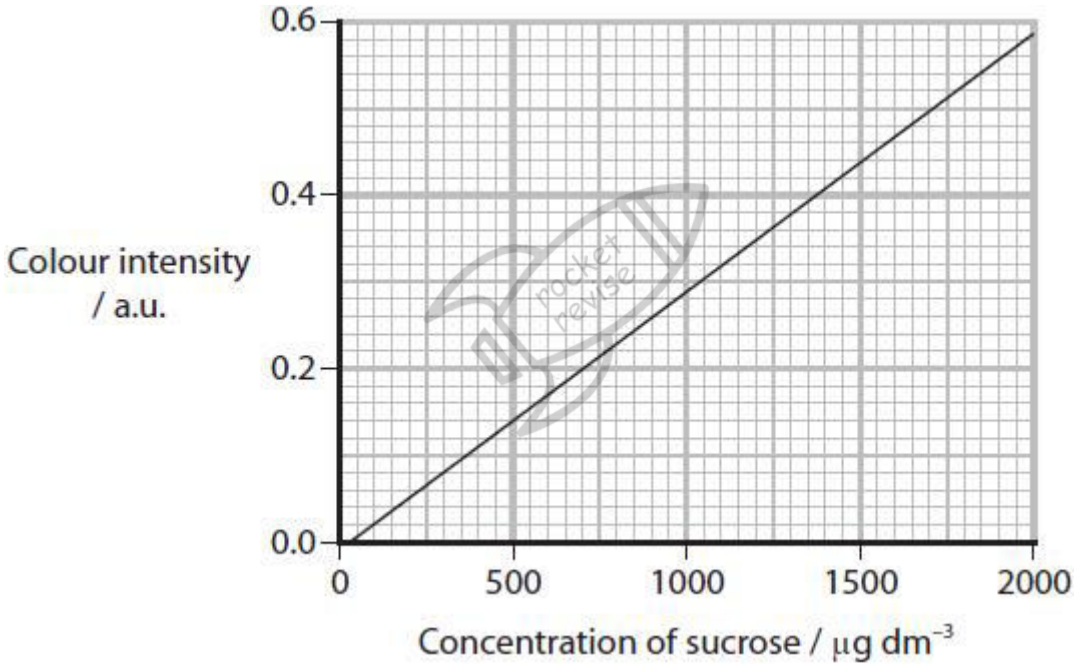
Add a chemical that binds to fructose and changes colour

↓

Measure the colour intensity of the solution

The colour intensity is directly proportional to the concentration of sucrose.
The graph shows this relationship.

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(i) Give a reason why the colour intensity is directly proportional to the concentration of sucrose.

(1)

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(ii) Suggest why the line does not start at the origin of the graph.

(1)

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(iii) State why the concentration of maltose and lactose cannot be measured using this test.

(1)

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(Total for question = 6 marks)

Q8.

Starch from plants is used in the production of food, paper and building materials.

The structural and chemical properties of starch depend on its amylose and amylopectin content.

(a) The table shows some of the bonds found in amylose and amylopectin.

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Put **one** cross ☒ in the appropriate box in each row to show whether the bond is present in these molecules.

(3)

Types of bond	Bonds found in			
	amylose only	amylopectin only	both amylose and amylopectin	neither amylose nor amylopectin
glycosidic bonds	☐	☐	☐	☐
1-6 α bonds	☐	☐	☐	☐
hydrogen bonds	☐	☐	☐	☐

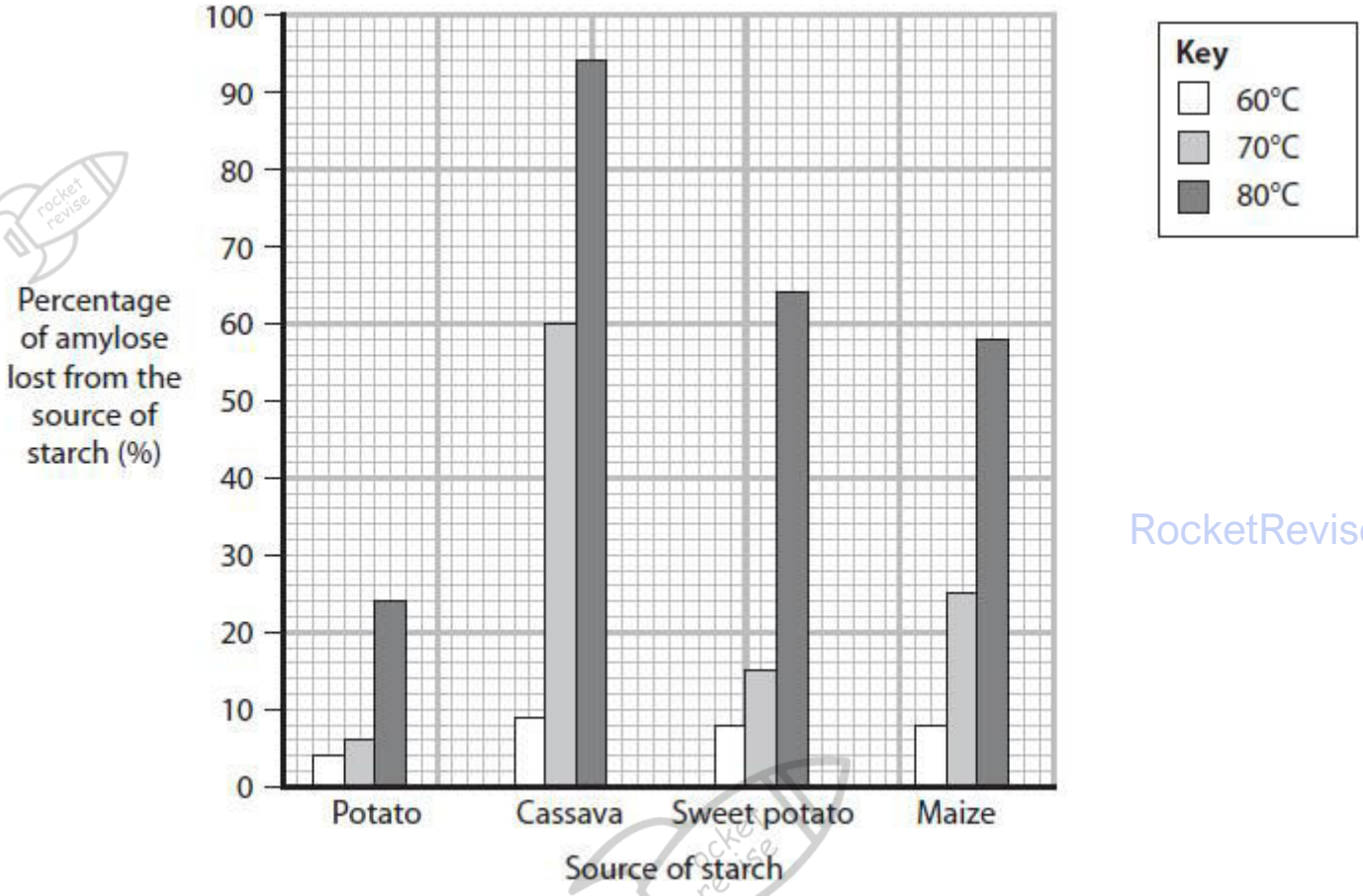
(b) When sources of starch are heated in water, amylose moves into the water.

The graph shows the percentage of amylose that is lost from four sources of starch at three temperatures.

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Describe **two** conclusions that can be made about the effects of temperature and source of starch on the loss of amylose.

(2)

1 Effect of temperature

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2 Effect of source of starch

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(c) The loss of amylose and amylopectin during the cooking of rice affects the hardness and stickiness of the cooked rice.

The effect of temperature on the loss of amylose and amylopectin was investigated.
The diagram shows part of the method used.

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Rice was mixed with water and soaked for 30 minutes

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This mixture was divided into nine equal samples

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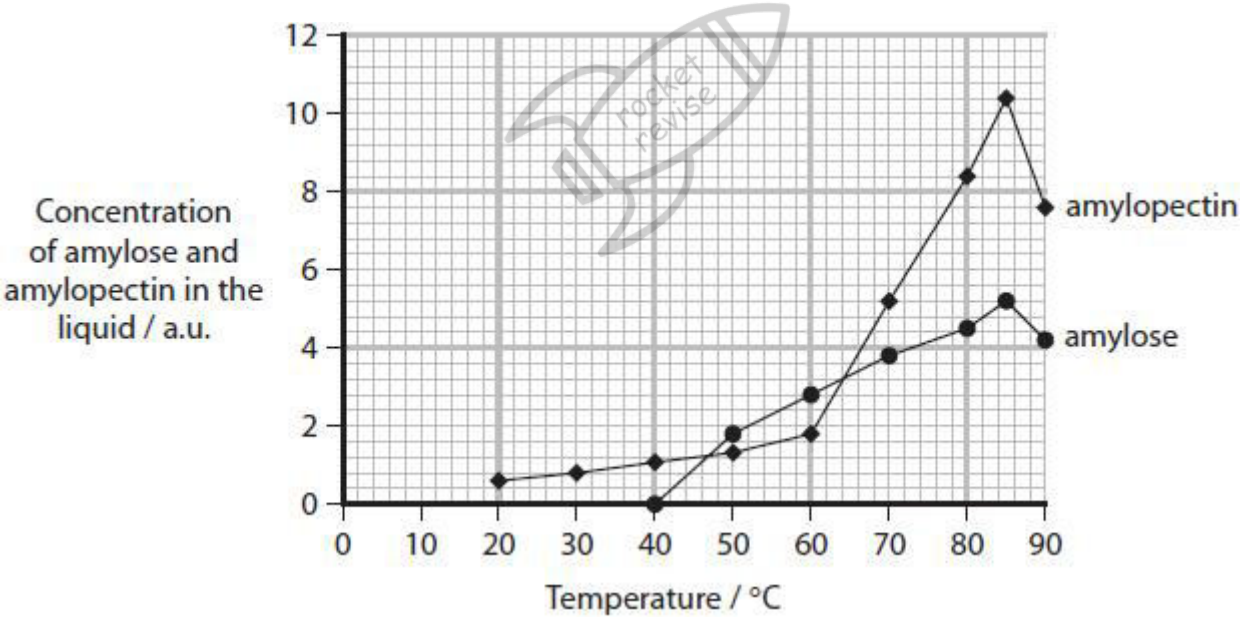
Each sample was heated to a different temperature and the liquid poured off when the temperature was reached

↓

The liquid collected at each temperature was analysed

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The graph shows the results of this investigation.



(i) Compare and contrast the effects of temperature on the loss of amylose and amylopectin from rice.

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(3)

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(ii) The investigation continued by adding a replacement liquid to finish cooking the rice.

The hardness and stickiness of the cooked rice was determined for each replacement liquid. The table shows the results of this investigation.

Replacement liquid	Hardness of cooked rice / a.u.	Stickiness of cooked rice / a.u.
Water	19.42	1.19
Water containing amylose	27.63	1.21
Water containing amylopectin	20.43	1.71
Water containing both amylose and amylopectin	27.65	1.70

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Comment on the effect of the replacement liquids on the hardness and stickiness of cooked rice. Use the information in the table to support your answer.

(3)

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(iii) Suggest why rice should be cooked in a measured volume of water, with the lid on, until all the cooking liquid has been absorbed.

Use the information in the table to support your answer.

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(2)

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(Total for question = 13 marks)

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Q9.

Water and carbohydrates are molecules found in all living organisms.

(a) Draw a diagram of a water molecule to show its dipole nature.

(2)

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(b) The table gives some statements about carbohydrates.

For each statement, put **one** cross in the appropriate box in each row to match the statement to the correct carbohydrates.

(3)

Statement	Carbohydrates			
	both monosaccharides and polysaccharides	**monosaccharides only**	**polysaccharides only**	**neither monosaccharides nor polysaccharides**
Have the general formula $C_nH_{2n}O_n$	☒	☒	☒	☒
Have glycosidic bonds	☒	☒	☒	☒
Have little effect on water potential	☒	☒	☒	☒

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(Total for question = 5 marks)

Q10.

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Lipids and carbohydrates can be used as energy storage molecules.

(a) Lactose is an important energy source in milk.

(i) What type of chemical reaction is involved in breaking a lactose molecule into two monosaccharides?

(1)

- ☒ **A** condensation
- ☒ **B** esterification
- ☒ **C** hydrogen bonding
- ☒ **D** hydrolysis

(ii) Milk contains 4.9 g of lactose per 100 g of milk.
Each gram of lactose provides 16 kJ of energy.
Calculate the energy available from the lactose in 200 g of milk.

(1)

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Answer ..... kJ

(b) Carbohydrates used as energy storage molecules include glycogen and starch.

Compare and contrast the structures of glycogen and starch.

(4)

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(c) (i) State one difference between the structure of a saturated lipid and an unsaturated lipid.

(1)

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(ii) Describe how a triglyceride is synthesised.

(3)

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(Total for question = 10 marks)

Q11.

Answer the questions with a cross in the boxes you think are correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Enzymes are biological catalysts.

(a) Enzyme molecules are formed from chains of amino acids joined together.

Which pair of elements is found in all amino acids?

(1)

- ☐ **A** carbon and oxygen
- ☐ **B** carbon and sulfur
- ☐ **C** nitrogen and magnesium
- ☐ **D** oxygen and magnesium

(b) Which row of the table describes how two amino acids are joined together in an enzyme molecule?

(1)

	Name of bond	Reaction that forms the bond
☐ **A**	ester	condensation
☐ **B**	ester	hydrolysis
☐ **C**	peptide	condensation
☐ **D**	peptide	hydrolysis

(c) Describe how a chain of amino acids can form an enzyme molecule.

(3)

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(Total for question = 5 marks)

Q12.

The primary structure of a protein determines its secondary structure and its three-dimensional structure.

(a) Read through the following account of the primary structure of a protein.

(5)

Complete the account by writing the most appropriate word or words on the dotted lines.
The primary structure of a protein is the specific sequence of amino acids joined

together by ..... bonds.

These bonds are formed between the ..... group of

one amino acid and the ..... group of an adjacent

amino acid by a ..... reaction.

These bonds are formed during the stage of protein synthesis

called ..... .

(b) The table describes the types of bond that hold the secondary and the three-dimensional structures together.

Which type of bonding is true for each structure?

(2)

Structure	Hydrogen bonds only	Ionic bonds only	Both hydrogen and ionic bonds	Neither of these bonds
secondary structure	☐	☐	☐	☐
three-dimensional structure	☐	☐	☐	☐

(Total for question = 7 marks)

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Q13.

Hair straightening, or rebonding, is a hair styling technique used to straighten hair using either heat or chemicals.

(a) Hair is made from an insoluble fibrous protein called keratin.

(i) Describe **two** structural features of insoluble fibrous proteins.

(2)

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2 .....

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(ii) Keratin contains a high proportion of the amino acid, cysteine.
Disulfide bridges form between the R groups of two cysteines.
The diagram shows the R group of cysteine.



Draw the complete structure of the amino acid, cysteine.

(3)

(b) The effect of two different temperatures on the lengths of the primary structure and secondary structure of a polypeptide was investigated.

The table shows the results of this investigation.

Structure of polypeptide	Length of structure / nm	
	at 25 °C	at 55 °C
primary	505	505
secondary	48	70

(i) Describe the secondary structure of a polypeptide.

(2)

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(ii) Explain the results of this investigation.

(4)

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(Total for question = 11 marks)

Q14.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

The polysaccharides and proteins that we eat are digested. The products of digestion are then transported in the blood plasma as a solution in water.

(a) Which of the following statements about water are correct?

1. A single molecule of water is composed of an oxygen atom joined to two hydrogen atoms by

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hydrogen bonds.
2. Water is a solvent.
3. The oxygen in a water molecule has a small negative charge and the hydrogen atoms each have a small positive charge.

(1)

- ☐ **A** 1 and 2 only
- ☐ **B** 1 and 3 only
- ☐ **C** 2 and 3 only
- ☐ **D** 1, 2 and 3

(b) The table shows some information about the structure of amylose and amylopectin.

Put one cross ☒ in the appropriate box in each row to show whether the structure is present in these molecules.

(3)

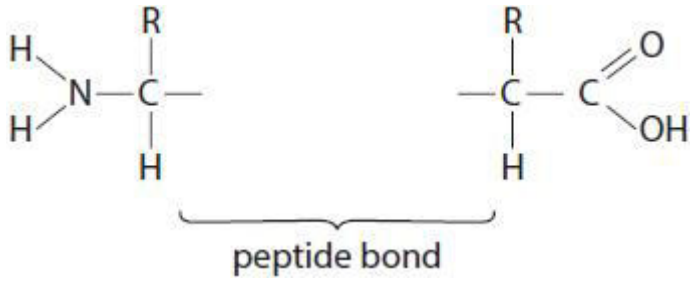
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Structure	Structure found in			
	amylose only	amylopectin only	both amylose and amylopectin	neither amylose nor amylopectin
glycosidic bonds	☐	☐	☐	☐
1-4 *α* bonds	☐	☐	☐	☐
branched side chains	☐	☐	☐	☐

(c) The diagram shows parts of two amino acids.

Complete the diagram to show a peptide bond joining these two amino acids.

(1)



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(Total for question = 5 marks)

Q15.

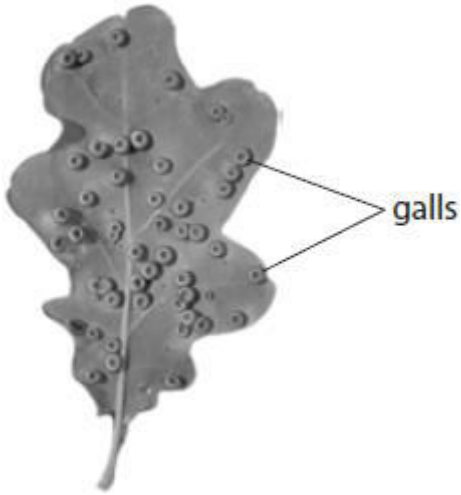
Some insects lay eggs inside leaves. This causes the leaves to produce swellings called galls.

The galls supply the developing insects with nutrients and protect them from the external environment.

The photograph shows galls on a leaf.

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In an investigation, the concentrations of protein molecules and amino acids found in healthy leaves, leaves with galls and in the galls themselves were determined.

(a) The insects were removed from each gall before the investigation.

Suggest why the insects were removed from the galls.

(2)

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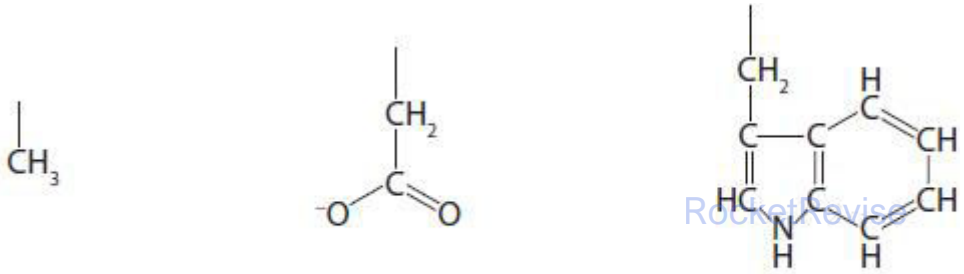
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(b) The amino acids were extracted and dissolved in a non-polar organic solvent.

(i) The diagram shows the R group of the amino acids, alanine, aspartate and tryptophan.



alanine aspartate tryptophan

Which row of the table describes the solubility of these amino acids in a non–polar organic solvent?

(1)

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	Solubility in a non-polar organic solvent Most soluble → Least soluble		
☐ A	alanine	aspartate	tryptophan
☐ B	alanine	tryptophan	aspartate
☐ C	aspartate	alanine	tryptophan
☐ D	aspartate	tryptophan	alanine

(ii) The solubility of an amino acid can be determined by measuring the maximum mass of the amino acid that dissolves in a known volume of solvent.

The solubility of the amino acid histidine in a solvent is 43.5 g dm^{-3}.

The mass of the amino acid leucine that dissolves in 250 cm^3 of the same solvent is 5.5 g.

Calculate how many times more soluble histidine is than leucine.

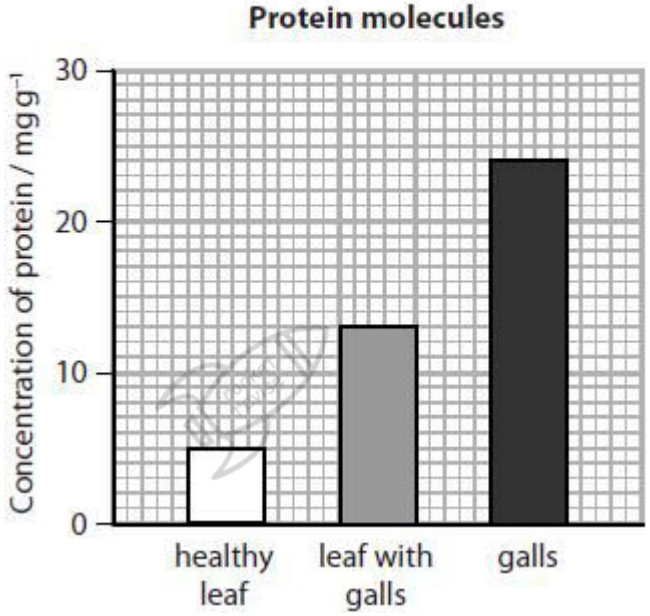
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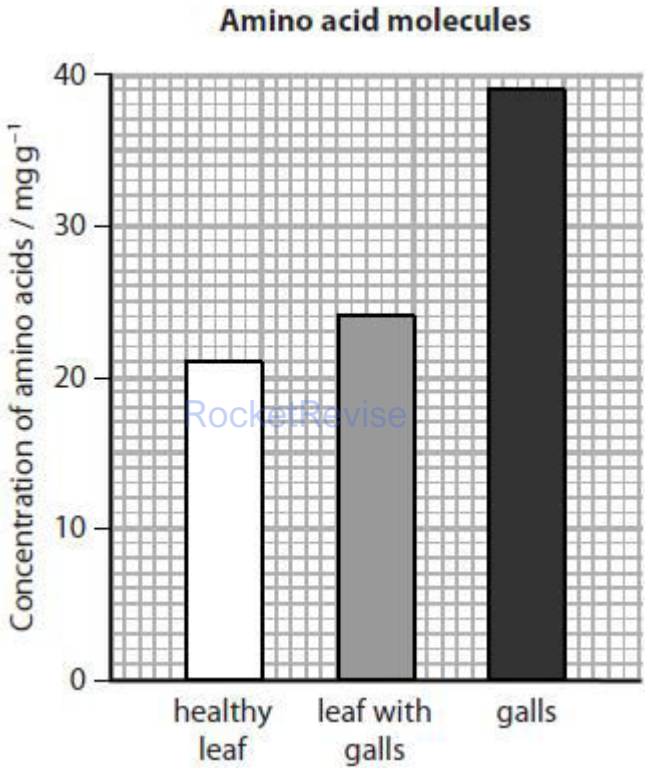
Answer .....

* (c) The gall-forming insects secrete saliva into the plant tissues.

The saliva contains enzymes that change the nutrients in the leaf and cause the galls to form.

The graphs show the concentrations of protein molecules and amino acid molecules in tissues from a healthy leaf, tissues from a leaf with galls and in the galls themselves.





The table shows the abundance of five amino acid molecules.

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Amino acid	Abundance in tissues of a healthy leaf	Abundance in tissues of a leaf with galls	Abundance in galls
alanine	++	+	++
arginine	++	–	++
histidine	–	–	++
leucine	+	+	+
tryptophan	–	–	–

Key

++ most abundant

– absent

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Explain the results of this investigation. Use the information in both graphs and the table to support your answer.

(6)

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(Total for question = 11 marks)

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